

Stresses, Strains, and Deflections in Flexible Pavements

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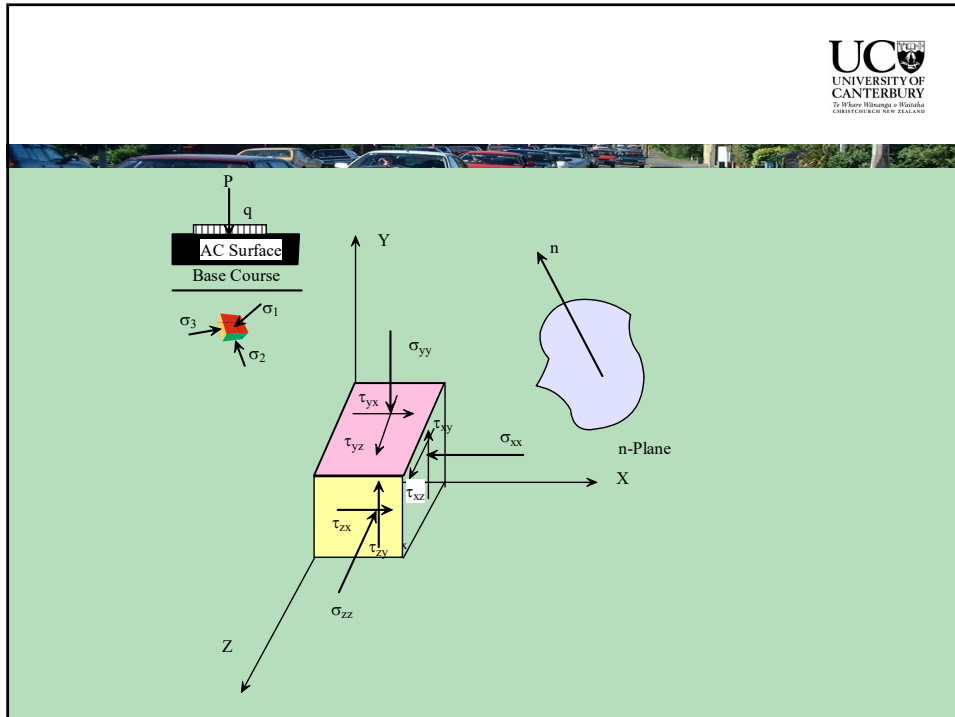
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Stresses and Strains in Flexible Pavements



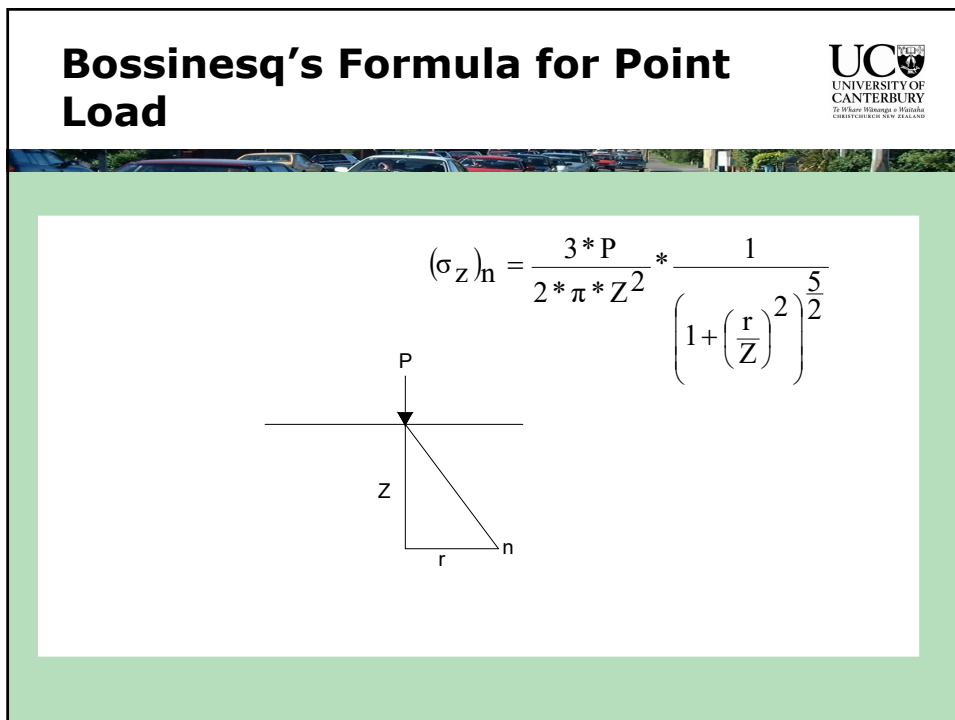
- Boussinesq's Equation (One layer)
- Assumptions:
 - 1. Homogeneous infinite half space
 - 2. Elastic and isotropic material

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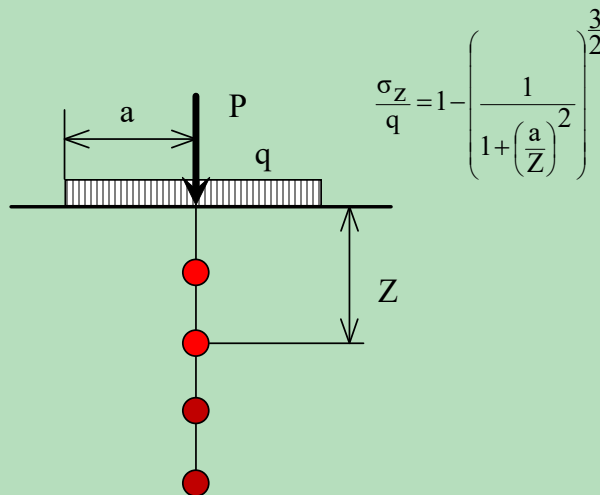
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Bossinesq's Formula for Point Load



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Vertical Stress Under the Centre of Circular Uniformly Loaded Area



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Solution of the Stresses and Strains by Charts

- Stresses
- Foster and Ahlvin developed charts
- Responses: normal stress σ_z , radial stress σ_r , tangential stress σ_t and deflection w . These charts were developed for a uniform circular load of intensity q over a circular area of radius a , and Poisson's ratio of 0.5.
- Strains
- After the stresses are calculated from charts these equations can be used to determine the strains

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Strains

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu^* (\sigma_r + \sigma_t)] \dots\dots\dots(1)$$

$$\epsilon_r = \frac{1}{E} [\sigma_r - \nu^* (\sigma_t + \sigma_z)] \dots\dots\dots(2)$$

$$\epsilon_t = \frac{1}{E} [\sigma_t - \nu^* (\sigma_z + \sigma_r)] \dots\dots\dots(3)$$

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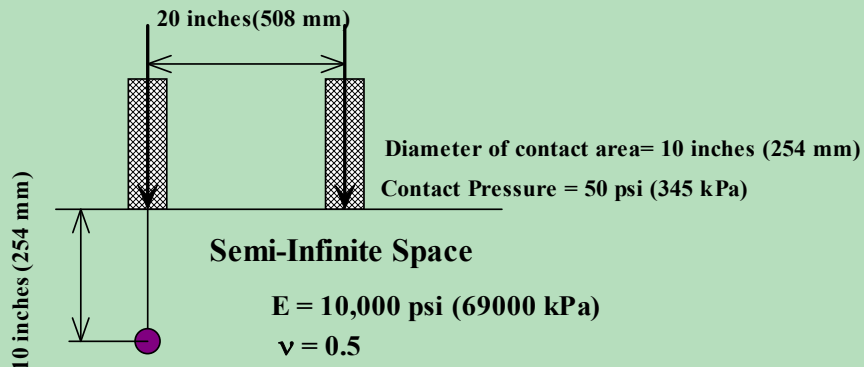
The Figure below shows homogeneous half space subjected to two circular loads, each 10 in (25.4 mm) in diameter and spaced at 20 in. (508 mm) on centres. The pressure on the circular area is 50 psi (345 kPa). The modulus of elasticity of the half space is 10000 psi (69 MPa) and Poisson ratio is 0.5.

Determine:

- 1. The vertical stress at point A
- 2. The vertical strain at Point A
- 3. The deflection at Point A
- Point A is located 10 in under the centre of one circle.

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Example



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	Load 1	Load 2	Total
Z/a	2.0	2.0	
r/a	0	4.0	
σ_z/q	28.33	0.75	29.08
σ_r/q	1.7	2.67	4.37
σ_t/q	1.7	0	1.7
F	0.667	0.208	0.875

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- Vertical Stress (σ_z) = $345 \times 29.08 / 100 = 100.33 \text{ kPa}$
- Radial Stress (σ_r) = $345 \times 4.37 / 100 = 15.08 \text{ kPa}$
- Tangential Stress (σ_t) = $345 \times 1.7 / 100 = 5.87 \text{ kPa}$
- Deflection (w) = $q \times a / E \times F = 345 \times 127 / 69000 \times 0.875 = 0.556 \text{ mm}$

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	Loads		$\sum_{i=1}^2 \text{Responses}$		
	P ₁	P ₂			
z/a	2	2	-----		
r/a	0	4	-----		
σ_z/q	28.33	0.75	29.08	14.54	σ_z
σ_r/q	1.7	2.67	4.37	2.185	σ_r
σ_t/q	1.7	0.0	1.7	0.85	σ_t
F	0.667	.208	.875	0.0219	W

$$\sigma_z = \sum_{i=1}^n \left(\frac{\sigma_z}{q} \right)_i * q_i$$

$$\sigma_r = \sum_{i=1}^n \left(\frac{\sigma_r}{q} \right)_i * q_i$$

$$\sigma_t = \sum_{i=1}^n \left(\frac{\sigma_t}{q} \right)_i * q_i$$

$$w = \sum_i F_i * \frac{q * a}{E}$$

$$\varepsilon_z = 1/10000 * [14.17 - 5(2.185 + .85)] = 0.00127$$

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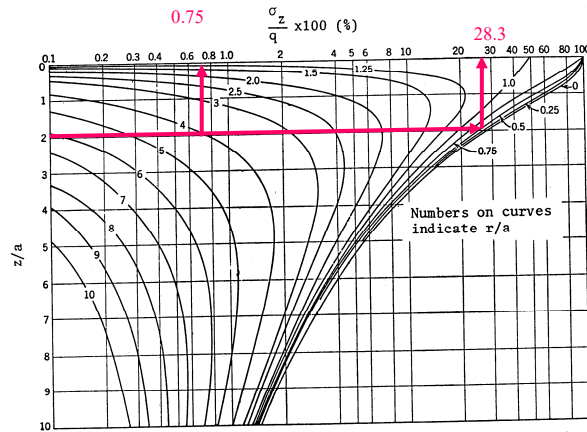


Figure 2.2 Vertical stresses due to circular loading. (After Foster and Ahlvin (1954).)

Sec. 2.1 Homogeneous Mass

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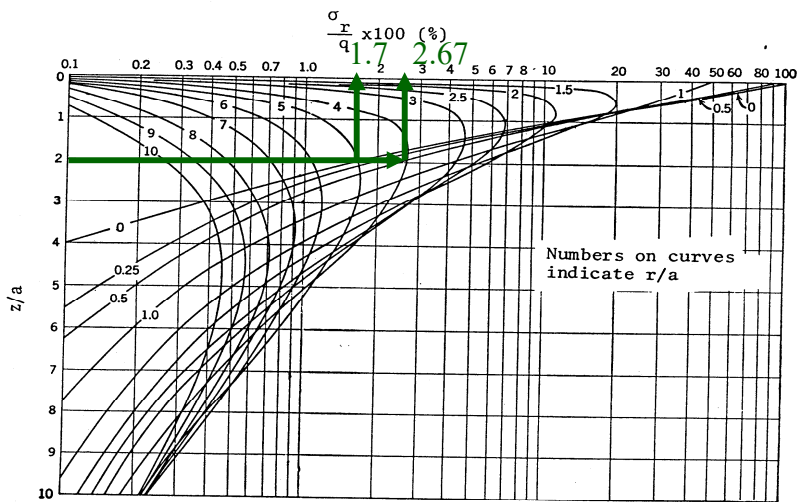
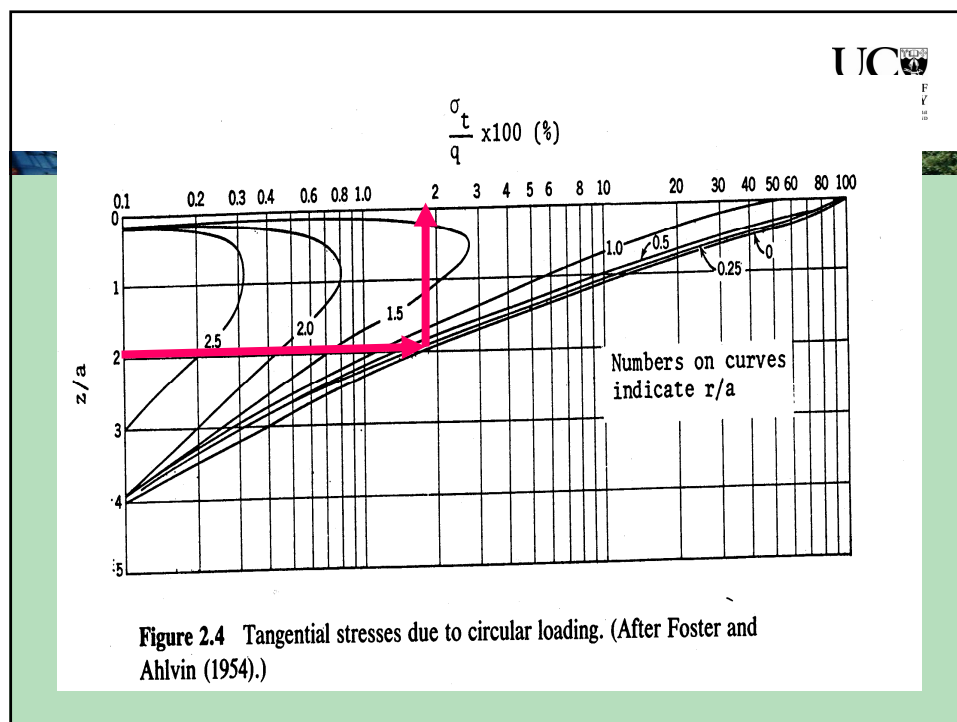
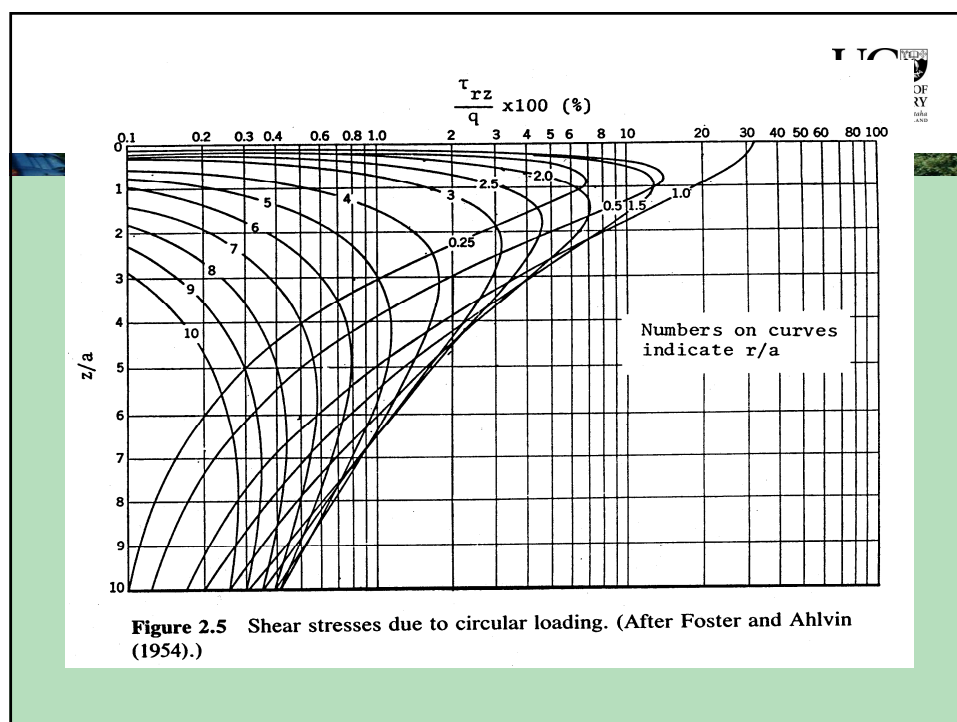


Figure 2.3 Radial stresses due to circular loading. (After Foster and Ahlvin, (1954).)

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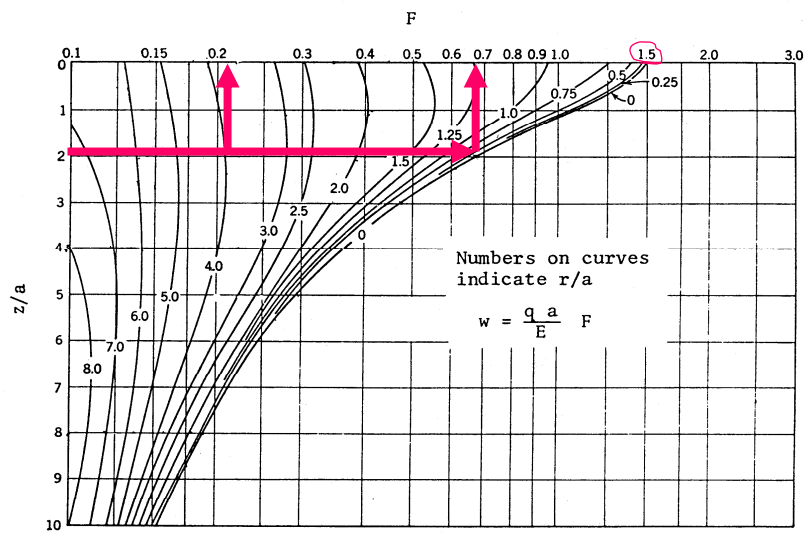
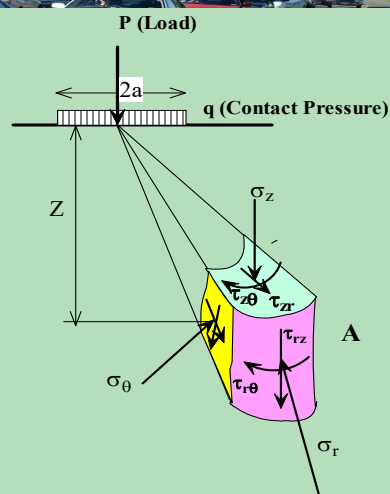


Figure 2.6 Vertical deflections due to circular loading. (After Foster and Ahlvin (1954).)

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Stresses under Axi-Symmetric Loading



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Solutions at the Axis of Symmetry

$$\sigma_z = q * \left[1 - \frac{z^3}{(a^2 + z^2)^{\frac{3}{2}}} \right] \dots\dots\dots(4)$$

$$\sigma_r = \sigma_t = \frac{q}{2} * \left[1 - 2\nu - \frac{2 * (1 + \nu) * z}{(a^2 + z^2)^{\frac{1}{2}}} + \frac{z^3}{(a^2 + z^2)^{\frac{3}{2}}} \right] \dots\dots\dots(5)$$

Strains:

$$\varepsilon_z = \frac{(1 + \nu) * q}{E} * \left[1 - 2\nu + \frac{2\nu * z}{(a^2 + z^2)^{\frac{1}{2}}} - \frac{z^3}{(a^2 + z^2)^{\frac{3}{2}}} \right] \dots\dots\dots(6)$$

$$\varepsilon_r = \frac{(1 + \nu) * q}{2E} * \left[1 - 2\nu - \frac{2 * (1 - \nu) * z}{(a^2 + z^2)^{\frac{1}{2}}} + \frac{z^3}{(a^2 + z^2)^{\frac{3}{2}}} \right] \dots\dots\dots(7)$$

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Vertical deflection:

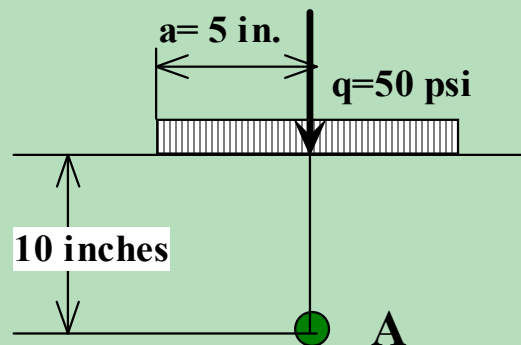
$$W = \frac{(1 + \nu)qa}{E} * \left\{ \frac{a}{(a^2 + z^2)^{\frac{1}{2}}} + \frac{(1 - 2 * \nu)}{a} * \left((a^2 + z^2)^{\frac{1}{2}} - z \right) \right\} \dots\dots\dots(8)$$

$$W = \frac{3 * q * a^2}{2 * E * (a^2 + z^2)^{\frac{1}{2}}} \quad \text{for } \nu = 0.5$$

$$w_o = \frac{2 * (1 - \nu^2) * q * a}{E} \quad \text{Surface Deflection}$$

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- Determine stresses, strains and deflections 10 inches under the centre of a uniformly loaded circular area of radius 5 inches and pressure 50 psi. Poisson's ratio 0.3.



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- $\sigma_z = 50 * (1 - 10^3 / (5^2 + 10^2)^{1.5}) = 14.22 \text{ psi}$
- $\sigma_r = \sigma_t = 50 / 2 * [1 + 2 * 0.3 - 2 * 1.3 * 10 / (5^2 + 10^2)^{0.5} + 10^3 / (5^2 + 10^2)^{1.5}] = -0.249 \text{ psi}$
- $\epsilon_z = 1.3 * 50 / 10000 * (1 - 0.6 + 6 / (5^2 + 10^2)^{0.5} - 10^3 / (5^2 + 10^2)^{1.5}) = 0.00144$
- $\epsilon_r = 1.3 * 50 / 20000 * [1 - 0.6 - 2 * 0.7 * 10 / (125)^{0.5} + 1000 / 125^{1.5}] = -0.000444$
- $w = 1.3 * 50 * 5 / 10000 * [5 / 125^{0.5} + (0.4 / 5) * (125^{0.5} - 10)] = 0.0176 \text{ in}$

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Rigid Plate

Pressure Distribution

$$q(r) = \frac{q * a}{2 * (a^2 - r^2)^{\frac{1}{2}}}$$

a = the radius of the rigid plate

q = the average pressure which is equal to the total load divided by the total area

r = is the distance from the centre of the plate to the point where the pressure is required.

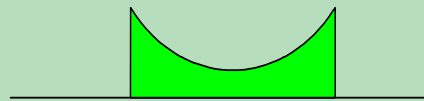
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Comparison between Flexible and Rigid Pavements

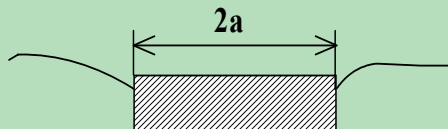
Uniform Pressure



Non-Uniform Pressure



Flexible Plate



Circular Rigid Plate

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Deflection in Rigid Pavements

$$W_o = \frac{\pi * (1 - \nu^2) * q * a}{2 * E}$$

Example

For the same radius, load and Poisson's ratio, compare the surface deflection under the centre for rigid and flexible plates.

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Solution

Surface Deflection for Flexible Pavements

$$W_{oFlex} = \frac{2 * (1 - \nu^2) * q * a}{E}$$

Surface Deflection for Rigid Pavements

$$W_{oRigid} = \frac{\pi * (1 - \nu^2) * q * a}{2 * E}$$

$$\frac{W_{oRigid}}{W_{oFlex}} = \frac{\pi * (1 - \nu^2) * q * a}{2 * E} * \frac{E}{2 * (1 - \nu^2) * q * a} * 100 = \frac{\pi}{4} * 100 = 78.5\%$$

This means that the deflection under the centre of the rigid pavements is 78.5% of that developed in the flexible pavements and this is attributed to the fact that the pressure at the centre of the rigid pavements is half the average value.

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Example

- In the plate loading test, the diameter of the plate was 40 cm and the load was 40 kN. The recorded deflection was 0.3 cm. Assuming Poisson's ratio equals 0.45 determine the modulus of elasticity of the subgrade.

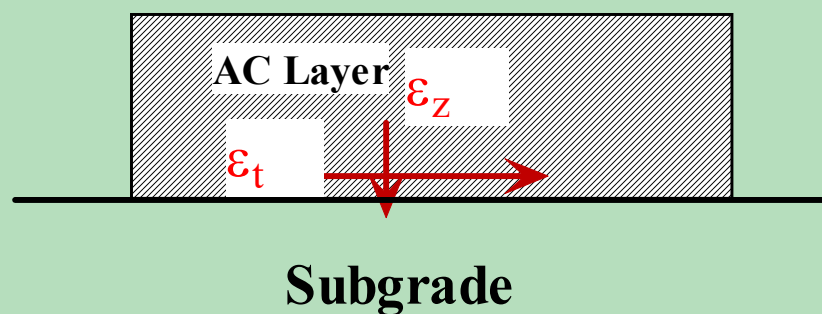
Solution

$$W_o = \frac{\pi * (1 - \nu^2) * q * a}{2 * E} = 0.003 = \frac{\pi * (1 - 0.45^2) * \frac{40}{\pi * 0.2^2} * 0.2}{2 * E} = \frac{79.75}{E}$$

$$E = 26.58 \text{ MPa}$$

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Layered Systems



$$\epsilon_t = \epsilon_{xx} \text{ or } \epsilon_{yy}$$

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Effect of Layers Moduli on Interface Stress



- The vertical stresses depends on the ratio of the layers moduli.
- For $E_1/E_2 = 1$ the vertical stresses at the interface is about 68% of the tyre pressure.
- For $E_1/E_2 = 100$, the vertical stress at the interface reduced dramatically to about 8%. This is expected because using stiff and strong pavement will dissipate the stresses significantly at the top of subgrade.

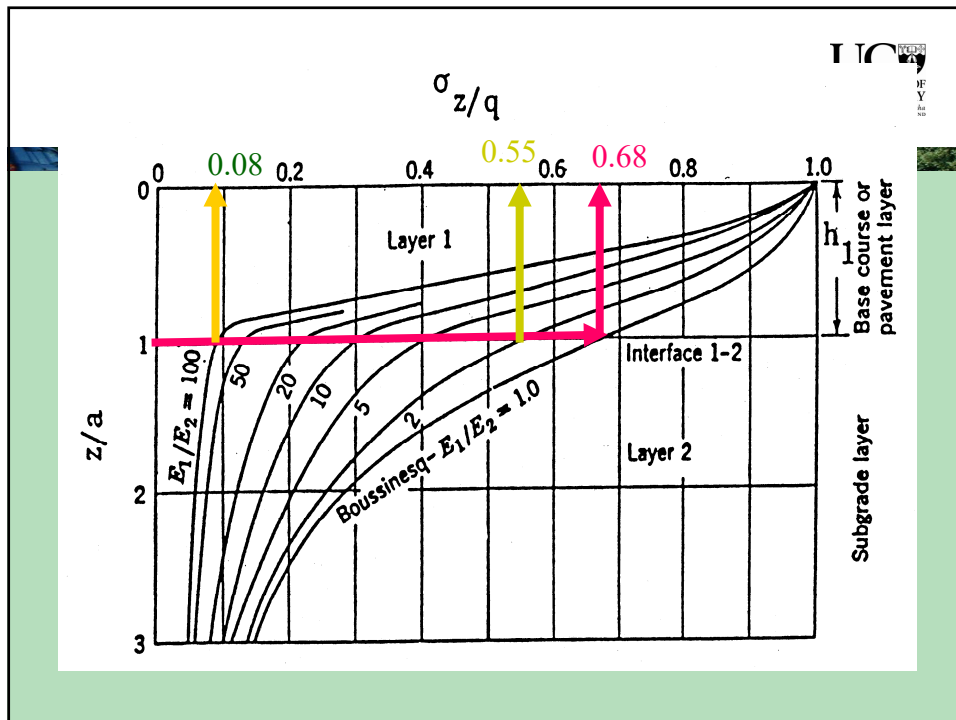
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Layers Moduli Effect Cont.

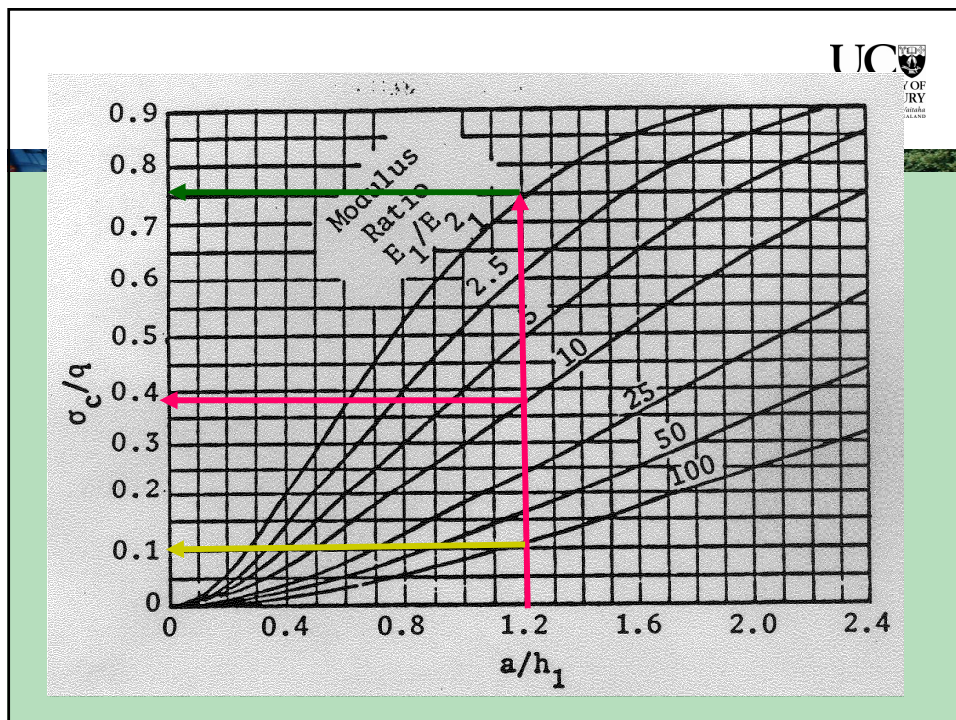


- The following Figure shows the combined effect of a/h and E_1/E_2 on the vertical stresses. The thicker the pavement the lower the vertical stresses at the interface. Also, the greater the loaded area, the higher the developed vertical stresses at the interface.

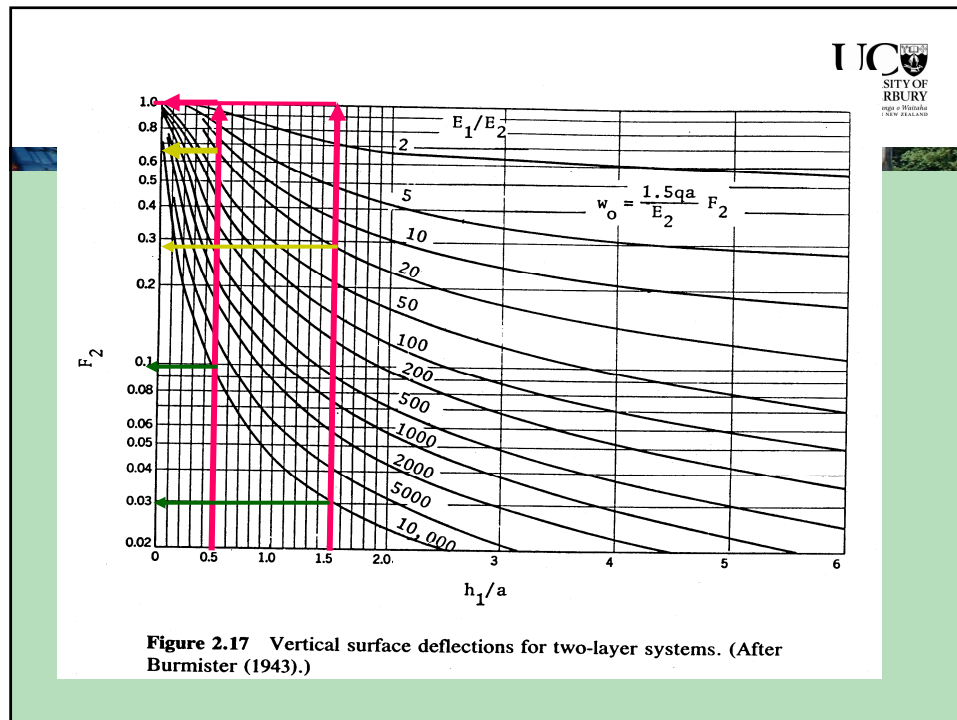
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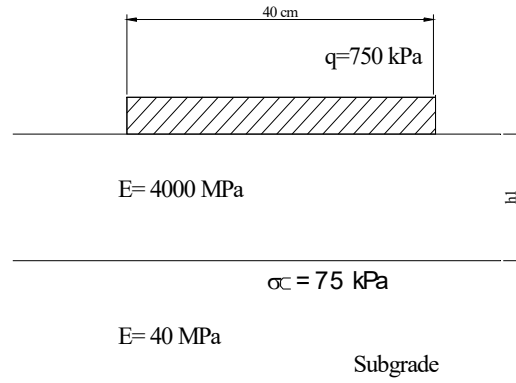


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▪ **Example:**

▪ A circular load with a radius of 20 cm and with a uniform pressure 750 kPa is applied on a two-layer system. The subgrade has an elastic modulus of 40 MPa and a maximum vertical stress at the interface is 75 kPa. If the HMA has an elastic modulus of 4.0 GPa, what is the required thickness of a full depth pavement? If a chip seal is applied on a granular base with an elastic modulus of 200 MPa, what is the thickness of the base course?

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Solution:

Case 1: Hot Mix Asphalt (HMA) over the subgrade

$$\frac{E_1}{E_2} = \frac{4000}{40} = 100$$

$$\frac{\sigma_c}{q} = \frac{75}{750} = 0.1$$

From Figure 2.15 $\frac{a}{h_1} = 1.10$ $a = 20 \text{ cm}$ then $h_1 = \frac{20}{1.10} = 18.2 \text{ cm}$

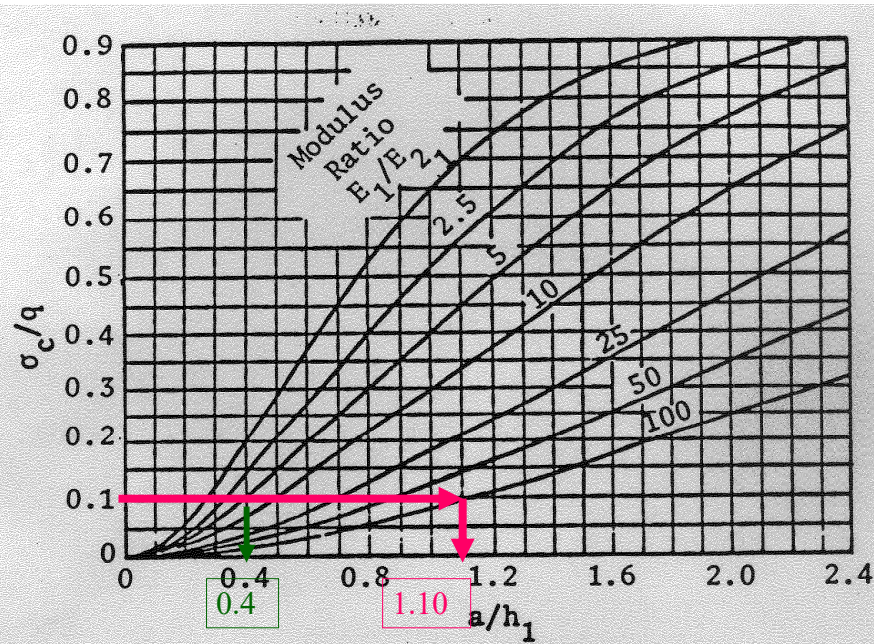
Case 2: Unbound base course over the subgrade

$$\frac{E_1}{E_2} = \frac{200}{40} = 5.0$$

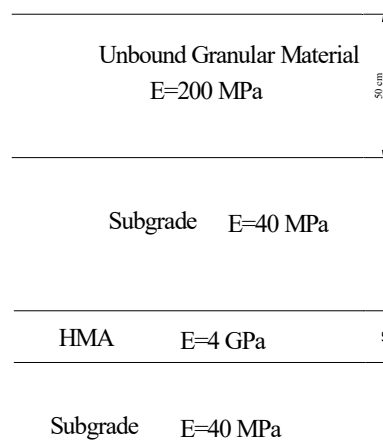
$$\frac{\sigma_c}{q} = \frac{75}{750} = 0.1$$

From Figure 2.15 $\frac{a}{h_1} = 0.4$ $a = 20 \text{ cm}$ then $h_1 = \frac{20}{0.4} = 50 \text{ cm}$

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- Example:
- A rigid plate test was applied on a two-layer system pavement. The total load on the plate was 90 kN and the diameter of the plate 30 cm. The top layer has a thickness of 20 cm and both layers have $\nu=0.5$. The elastic modulus of the bottom layer is 39 MPa. If the measured deflection is 3 mm determine the modulus of the top layer.

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Solution:

$$W_0 = \frac{1.18 * q * a}{E_2} * F_2 \dots\dots\dots$$

$$P=90 \text{ kN} \quad a=15 \text{ cm}, \quad h=20 \text{ cm} \quad E_2=39 \text{ MPa}$$

$$q = \frac{90}{\pi * 0.15^2} = 1273 \text{ kPa}$$

$$\frac{3}{1000} = \frac{1.18 * 1273 * 1000 * 0.15}{39 * 10^6} * F_2$$

$$F_2 = 0.519$$

$$\frac{h_1}{a} = \frac{20}{15} = 1.33$$

From Figure 2.17

$$\frac{E_1}{E_2} = 5$$

$$E_1 = 5.0 * E_2 = 5.0 * 39 = 195 \text{ MPa}$$

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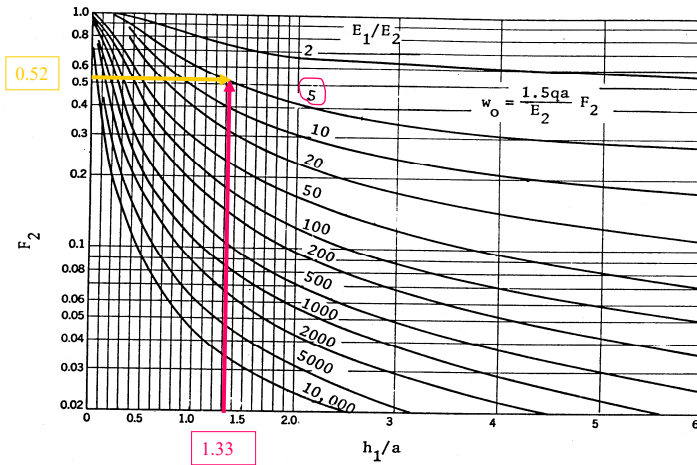
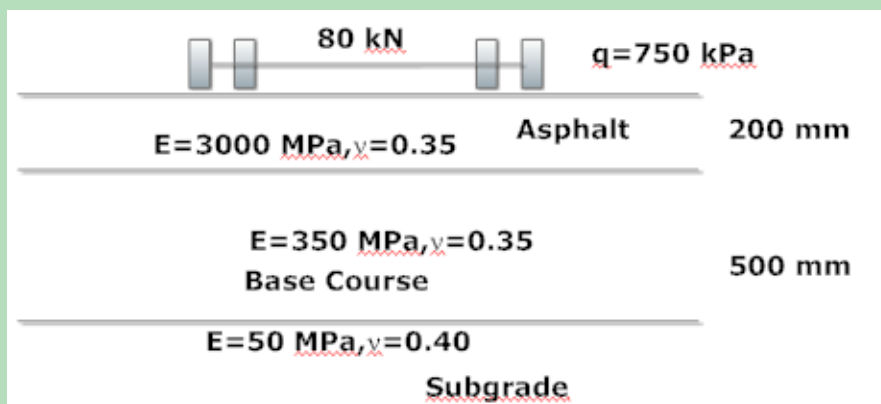


Figure 2.17 Vertical surface deflections for two-layer systems. (After Burmister (1943).)

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Circly Analysis



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